

2. A parcel of mass 5 kg is projected with speed 8 m s^{-1} up a line of greatest slope of a fixed rough inclined ramp.

The ramp is inclined at angle α to the horizontal, where $\sin \alpha = \frac{1}{7}$

The parcel is projected from the point A on the ramp and comes to instantaneous rest at the point B on the ramp, where $AB = 14 \text{ m}$.

The coefficient of friction between the parcel and the ramp is μ .

In a model of the parcel's motion, the parcel is treated as a particle.

- (a) Use the work-energy principle to find the value of μ .

(5)

- (b) Suggest one way in which the model could be refined to make it more realistic.

(1)

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2. A rough plane is inclined to the horizontal at an angle θ , where $\tan \theta = \frac{3}{4}$

A particle P of mass m is at rest at a point on the plane.

The particle is projected **up** the plane with speed $\sqrt{2ag}$

The particle moves up a line of greatest slope of the plane and comes to instantaneous rest after moving a distance d .

The coefficient of friction between P and the plane is $\frac{1}{7}$

- (a) Show that the magnitude of the frictional force acting on P as it moves up the plane

is $\frac{4mg}{35}$ (3)

Air resistance is assumed to be negligible.

Using the work-energy principle,

- (b) find d in terms of a . (4)

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2. A car of mass 1000 kg moves in a straight line along a horizontal road at a constant speed $U \text{ m s}^{-1}$. The resistance to the motion of the car is a constant force of magnitude 400 N.

The engine of the car is working at a constant rate of 16 kW.

- (a) Find the value of U .

(3)

The car now pulls a trailer of mass 600 kg in a straight line along the road using a tow rope which is parallel to the direction of motion. The resistance to the motion of the car is again a constant force of magnitude 400 N. The resistance to the motion of the trailer is a constant force of magnitude 300 N.

The engine of the car is working at a constant rate of 16 kW.

The tow rope is modelled as being light and inextensible.

Using the model,

- (b) find the tension in the tow rope at the instant when the speed of the car is $\frac{20}{3} \text{ m s}^{-1}$

(5)

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1. A van of mass 900 kg is moving along a straight horizontal road.

At the instant when the speed of the van is $v \text{ m s}^{-1}$, the resistance to the motion of the van is modelled as a force of magnitude $(500 + 7v) \text{ N}$.

When the engine of the van is working at a constant rate of 18 kW, the van is moving along the road at a constant speed $V \text{ m s}^{-1}$

- (a) Find the value of V .

(5)

Later on, the van is moving up a straight road that is inclined to the horizontal at an angle θ , where $\sin \theta = \frac{1}{21}$

At the instant when the speed of the van is $v \text{ m s}^{-1}$, the resistance to the motion of the van from non-gravitational forces is modelled as a force of magnitude $(500 + 7v) \text{ N}$.

The engine of the van is again working at a constant rate of 18 kW.

- (b) Find the acceleration of the van at the instant when $v = 15$

(4)

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2. A truck of mass 1200 kg is moving along a straight horizontal road.

At the instant when the speed of the truck is $v \text{ ms}^{-1}$, the resistance to the motion of the truck is modelled as a force of magnitude $(900 + 9v) \text{ N}$.

The engine of the truck is working at a constant rate of 25 kW.

- (a) Find the deceleration of the truck at the instant when $v = 25$

(4)

Later on, the truck is moving up a straight road that is inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{20}$

At the instant when the speed of the truck is $v \text{ ms}^{-1}$, the resistance to the motion of the truck from non-gravitational forces is modelled as a force of magnitude $(900 + 9v) \text{ N}$.

When the engine of the truck is working at a constant rate of 25 kW the truck is moving up the road at a constant speed of $V \text{ ms}^{-1}$.

- (b) Find the value of V .

(5)

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5. A car of mass 600 kg is moving along a straight horizontal road.

At the instant when the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car is modelled as a force of magnitude $(200 + 2v) \text{ N}$.

The engine of the car is working at a constant rate of 12 kW.

- (a) Find the acceleration of the car at the instant when $v = 20$

(4)

Later on the car is moving up a straight road inclined at an angle θ to the horizontal,

where $\sin \theta = \frac{1}{14}$

At the instant when the speed of the car is $v \text{ m s}^{-1}$, the resistance to the motion of the car from non-gravitational forces is modelled as a force of magnitude $(200 + 2v) \text{ N}$.

The engine is again working at a constant rate of 12 kW.

At the instant when the car has speed $w \text{ m s}^{-1}$, the car is decelerating at 0.05 m s^{-2} .

- (b) Find the value of w .

(5)

5. A light elastic string has natural length $2a$ and modulus of elasticity $2mg$. One end of the string is attached to a fixed point A on a horizontal ceiling. The other end is attached to a particle P of mass m .

The particle P hangs in equilibrium at the point E , where $AE = 3a$.

The particle P is then projected vertically downwards from E with speed $\frac{3}{2}\sqrt{ag}$

Air resistance is assumed to be negligible.

Find the elastic energy stored in the string, when P first comes to instantaneous rest. Give your answer in the form $kmg a$, where k is a constant to be found.

(7)

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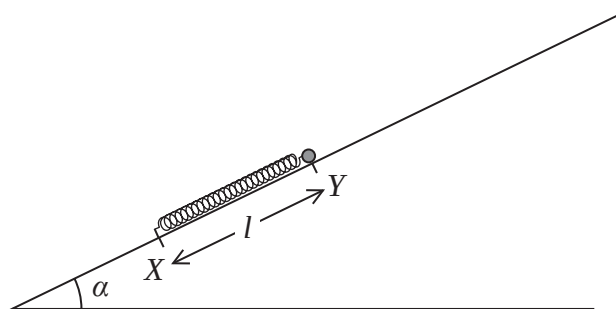


Figure 2

A light elastic spring has natural length $3l$ and modulus of elasticity $3mg$.

One end of the spring is attached to a fixed point X on a rough inclined plane.

The other end of the spring is attached to a package P of mass m .

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The package is initially held at the point Y on the plane, where $XY = l$. The point Y is higher than X and XY is a line of greatest slope of the plane, as shown in Figure 2.

The package is released from rest at Y and moves up the plane.

The coefficient of friction between P and the plane is $\frac{1}{3}$

By modelling P as a particle,

(a) show that the acceleration of P at the instant when P is released from rest is $\frac{17}{15}g$ (5)

(b) find, in terms of g and l , the speed of P at the instant when the spring first reaches its natural length of $3l$. (6)

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6. A light elastic string with natural length l and modulus of elasticity kmg has one end attached to a fixed point A on a rough inclined plane. The other end of the string is attached to a package of mass m .

The plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{5}{12}$

The package is initially held at A . The package is then projected with speed $\sqrt{6gl}$ up a line of greatest slope of the plane and first comes to rest at the point B , where $AB = 3l$.

The coefficient of friction between the package and the plane is $\frac{1}{4}$

By modelling the package as a particle,

- (a) show that $k = \frac{15}{26}$ (6)
- (b) find the acceleration of the package at the instant it starts to move back down the plane from the point B . (5)

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4. A car of mass 600 kg pulls a trailer of mass 150 kg along a straight horizontal road. The trailer is connected to the car by a light inextensible towbar, which is parallel to the direction of motion of the car. The resistance to the motion of the trailer is modelled as a constant force of magnitude 200 N. At the instant when the speed of the car is $v \text{ ms}^{-1}$, the resistance to the motion of the car is modelled as a force of magnitude $(200 + \lambda v) \text{ N}$, where λ is a constant.

When the engine of the car is working at a constant rate of 15 kW, the car is moving at a constant speed of 25 ms^{-1}

- (a) Show that $\lambda = 8$

(4)

Later on, the car is pulling the trailer up a straight road inclined at an angle θ to the horizontal, where $\sin \theta = \frac{1}{15}$

The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 200 N at all times. At the instant when the speed of the car is $v \text{ ms}^{-1}$, the resistance to the motion of the car from non-gravitational forces is modelled as a force of magnitude $(200 + 8v) \text{ N}$.

The engine of the car is again working at a constant rate of 15 kW.

When $v = 10$, the towbar breaks. The trailer comes to instantaneous rest after moving a distance d metres up the road from the point where the towbar broke.

- (b) Find the acceleration of the car immediately after the towbar breaks.

(4)

- (c) Use the work-energy principle to find the value of d .

(4)



7. A spring of natural length a has one end attached to a fixed point A . The other end of the spring is attached to a package P of mass m .
The package P is held at rest at the point B , which is vertically below A such that $AB = 3a$.

After being released from rest at B , the package P first comes to instantaneous rest at A .
Air resistance is modelled as being negligible.

By modelling the spring as being light and modelling P as a particle,

- (a) show that the modulus of elasticity of the spring is $2mg$ (5)

- (b) (i) Show that P attains its maximum speed when the extension of the spring is $\frac{1}{2}a$

- (ii) Use the principle of conservation of mechanical energy to find the maximum speed, giving your answer in terms of a and g . (6)

In reality, the spring is not light.

- (c) State one way in which this would affect your energy equation in part (b). (1)

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7. A particle P , of mass m , is attached to one end of a light elastic spring of natural length a and modulus of elasticity kmg .

The other end of the spring is attached to a fixed point O on a ceiling.

The point A is vertically below O such that $OA = 3a$

The point B is vertically below O such that $OB = \frac{1}{2}a$

The particle is held at rest at A , then released and first comes to instantaneous rest at the point B .

- (a) Show that $k = \frac{4}{3}$ (3)
- (b) Find, in terms of g , the acceleration of P immediately after it is released from rest at A . (3)
- (c) Find, in terms of g and a , the maximum speed attained by P as it moves from A to B . (6)

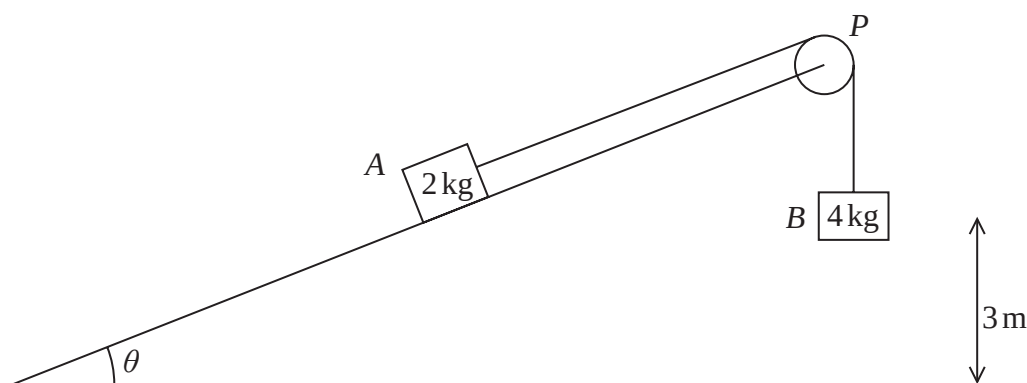
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6.

**Figure 4**

Two blocks, *A* and *B*, of masses 2 kg and 4 kg respectively are attached to the ends of a light inextensible string.

Initially *A* is held on a fixed rough plane. The plane is inclined to horizontal ground at an angle θ , where $\tan \theta = \frac{3}{4}$

The string passes over a small smooth light pulley *P* that is fixed at the top of the plane. The part of the string from *A* to *P* is parallel to a line of greatest slope of the plane.

Block *A* is held on the plane with the distance *AP* greater than 3 m.

Block *B* hangs freely below *P* at a distance of 3 m above the ground, as shown in Figure 4.

The coefficient of friction between *A* and the plane is μ

Block *A* is released from rest with the string taut.

By modelling the blocks as particles,

- (a) find the potential energy lost by the whole system as a result of *B* falling 3 m. (3)

Given that the speed of *B* at the instant it hits the ground is 4.5 m s^{-1} and ignoring air resistance,

- (b) use the work-energy principle to find the value of μ (6)

After *B* hits the ground, *A* continues to move up the plane but does not reach the pulley in the subsequent motion.

Block *A* comes to instantaneous rest after moving a total distance of $(3 + d)$ m from its point of release.

Ignoring air resistance,

- (c) use the work-energy principle to find the value of d (4)

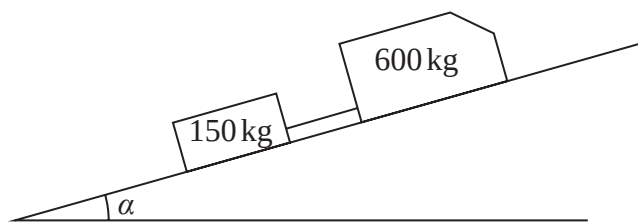
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2.

**Figure 1**

A van of mass 600 kg is moving up a straight road which is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{15}$. The van is towing a trailer of mass 150 kg. The van is attached to the trailer by a towbar which is parallel to the direction of motion of the van and the trailer, as shown in Figure 1.

The resistance to the motion of the van from non-gravitational forces is modelled as a constant force of magnitude 200 N.

The resistance to the motion of the trailer from non-gravitational forces is modelled as a constant force of magnitude 100 N.

The towbar is modelled as a light rod.

The engine of the van is working at a constant rate of 12 kW.

Find the tension in the towbar at the instant when the speed of the van is 9 m s^{-1}

(8)

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7. A particle P of mass m is attached to one end of a light elastic string of natural length a and modulus of elasticity $3mg$.

The other end of the string is attached to a fixed point O on a ceiling.

The particle hangs freely in equilibrium at a distance d vertically below O .

- (a) Show that $d = \frac{4}{3}a$. (3)

The point A is vertically below O such that $OA = 2a$.

The particle is held at rest at A , then released and first comes to instantaneous rest at the point B .

- (b) Find, in terms of g , the acceleration of P immediately after it is released from rest. (3)

- (c) Find, in terms of g and a , the maximum speed attained by P as it moves from A to B . (5)

- (d) Find, in terms of a , the distance OB . (3)

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